

8(a)	<i>two from:</i> 1 (two or more) genes on same chromosome ; 2 not sex chromosome ; 3 will be inherited together ;	2
8(b)	<i>four from:</i> 1 most offspring are, purple long and red round / parental phenotypes ; 2 fewer offspring are, purple round and red long / non-parental phenotypes ; 3 recombinants / new combinations of alleles ; 4 (because of) crossing over ; 5 between the two gene loci ; 6 AVP ; e.g. COV is 12% / low COV / two genes not very far apart	4

Question	Answer	Marks
8(c)	<i>three from:</i> 1 change in, base (pair) / nucleotide, sequence in <u>DNA</u> ; 2 random / spontaneous ; <i>due to</i> 3 <i>ref. to</i> base, substitution / deletion / insertion ; 4 (during) DNA replication ; 5 failure of proof reading mechanism ; 6 <i>ref. to</i> mutagenic agents ; e.g. ionising radiation / UV radiation	3